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March 15, 2022
NRC-22-0008

10 CFR 50.73

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, DC 20555-0001

Fermi 2 Power Plant
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2021-002-01

Pursuant to 10 CFR 50.73(a)(2)(v)(D), DTE Electric Company (DTE) is submitting revision 01 to LER No. 2021-002, Unplanned Inoperability of High Pressure Coolant Injection System due to an Inverter Circuit Failure.

No new commitments are being made in this submittal.

Should you have any questions or require additional information, please contact Mr. Ertman L. Bennett III, Manager – Nuclear Licensing, at (734) 586-4273.

Sincerely,

A handwritten signature in black ink, appearing to be "P. Dietrich", written over a large, stylized circular flourish.

Peter Dietrich
Senior Vice President and Chief Nuclear Officer

Enclosure: Licensee Event Report No. 2021-002-01, Unplanned Inoperability of High Pressure Coolant Injection System Due to an Inverter Circuit Failure

cc: NRC Project Manager
NRC Resident Office
Regional Administrator, Region III

**Enclosure to
NRC-22-0008**

**Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43**

**Licensee Event Report (LER) No. 2021-002-01
Unplanned Inoperability of High Pressure Coolant Injection System Due to an Inverter
Circuit Failure**



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollcts.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk at: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Fermi 2	2. Docket Number 05000 341	3. Page 1 OF 5
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4. Title
Unplanned Inoperability of High Pressure Coolant Injection System Due to an Inverter Circuit Failure

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
8	22	2021	2021	- 002 -	1	03	15	2022	N/A	05000
									Facility Name	Docket Number
									N/A	05000
9. Operating Mode 1						10. Power Level 100				

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	
☐ OTHER (Specify here, in abstract, or NRC 366A).				

12. Licensee Contact for this LER

Licensee Contact Ertman L. Bennett III – Manager, Nuclear Licensing	Phone Number (Include area code) 734-586-4273
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	BJ	INVT	X999	Yes	X	DG	34	X999	Yes

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

At 0529 Eastern Daylight Savings Time (EDT) on August 22, 2021, while in Mode 1 at 100% power, alarms occurred for the Division 2 emergency diesel generator (EDG) load sequencer and the High Pressure Coolant Injection (HPCI) inverter circuit. These issues led to Division 2 EDGs and HPCI being declared inoperable per Technical Specification (TS) 3.8.1 and TS 3.5.1, respectively. Subsequent failures of the Division 2 battery charger and inoperability of the Division 2 direct current (DC) electrical power distribution subsystem were identified under TS 3.8.4 and TS 3.8.7, respectively. The cause of the event was a voltage spike of unknown origin. However, the spare battery charger was placed in service for Division 2 and the replacement of the power supplies for the load sequencer and HPCI inverter eliminated the problem and allowed for all the systems to be returned to service with no further issues by (or before) 0956 EDT on August 24, 2021. Other safety systems such as the Reactor Core Isolation Cooling (RCIC) system and Division 1 EDGs remained operable throughout the event. Although no violations of TS occurred, the unplanned inoperability of HPCI is a reportable condition. The safety significance of this event was determined to be very low and there were no radiological releases.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME		2. DOCKET NUMBER		3. LER NUMBER		
Fermi 2		05000-	341	YEAR 2021	SEQUENTIAL NUMBER 002	REV NO. 1

NARRATIVE**INITIAL PLANT CONDITIONS**

Mode – 1
Reactor Power – 100%

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

At 0529 Eastern Daylight Savings Time (EDT) on August 22, 2021, alarms [ALM] occurred for the Division 2 emergency diesel generator (EDG) [DG] load sequencer [34] and the High Pressure Coolant Injection (HPCI) [BJ] inverter circuit. These issues led to Division 2 EDGs and HPCI being declared inoperable per TS 3.8.1, "Alternating Current (AC) Sources - Operating," Required Action B.4 and TS 3.5.1, "Emergency Core Cooling Systems (ECCS) – Operating," Condition E.1, respectively. Offsite power circuits and Division 1 EDGs were verified to be operable in accordance with TS 3.8.1 Required Actions B.1 and B.3.1, respectively. The Reactor Core Isolation Cooling (RCIC) [BN] system was verified operable in accordance with TS 3.5.1 Required Action E.1.

The HPCI system and the EDG load sequencer are powered by the Division 2 battery. The initial investigation identified blown fuses associated with the power supplies of both the Division 2 EDG load sequencer and HPCI. The fuses were replaced but blew again. The blown replacement fuses were removed and additional troubleshooting activities were performed. During the subsequent troubleshooting activities, fluctuations were identified in the Division 2 battery charger [BYC] in the direct current (DC) electrical power subsystem [EI]. The Division 2 battery charger was declared inoperable at 2233 EDT on August 22, 2021, and TS 3.8.4, "DC Sources – Operating," Condition A was entered. Shortly thereafter, at 2331 EDT, the Division 2 DC electrical power distribution subsystem was also declared inoperable due to low voltage and TS 3.8.7, "Distribution Systems – Operating," Condition B was entered. The spare battery charger was placed in service for Division 2. Using the spare, the Division 2 battery charger was restored to an operable status at 0046 EDT on August 23, 2021 and TS 3.8.4 Condition A was exited. Placement of the spare battery charger also stabilized voltage on the Division 2 DC electrical power distribution subsystem such that TS 3.8.7 Condition B was exited at 0055 EDT on August 23, 2021. The durations of the inoperability's of the Division 2 battery charger and DC electrical power distribution subsystem were approximately 2 hours and 1 hour, respectively which were both within the TS allowable durations of 4 hours and 2 hours, respectively. No other safety-related systems were out of service during the event.

Following transfer to the Division 2 spare battery charger, the aux transformer was replaced in the original battery charger, the power supplies for the Division 2 EDG load sequencer and HPCI Inverter were replaced. The Division 2 EDG load sequencer was restored to service and the EDGs were declared operable at 2132 EDT on August 23, 2021. The duration of the Division 2 EDG load sequencer inoperability was approximately 40 hours which was within the TS 3.8.1 Condition B

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allowance of 72 hours since offsite power sources were available. HPCI was restored to service and declared operable at 0956 EDT on August 24, 2021. The duration of the HPCI inoperability was approximately 52 hours which was within the TS 3.5.1 Condition E allowance of 14 days since RCIC was operable. A review verified that the offsite power sources, Division 1 EDGs, and RCIC were operable during the entire event. All TS required actions were taken within the allowable times and no violation of TS occurred.

An 8-hour event notification (EN 55423) was made to the NRC on August 22, 2021 based on meeting the reporting criteria of Title 10 Code of Federal Regulations (10 CFR) 50.72(b)(3)(v)(D) since an unplanned HPCI inoperability is an event or condition that at the time of discovery could have prevented the fulfillment of the safety function of a system needed to mitigate the consequences of an accident. This LER is being made under the corresponding criteria of 10 CFR 50.73(a)(2)(v)(D); specifically, that the 52-hour unplanned inoperability of the HPCI system was an event or condition that could have prevented the fulfillment of the safety function of a system needed to mitigate the consequences of an accident.

No reportability criteria were met for the EDGs, Division 2 battery charger, or Division 2 DC electrical power distribution subsystem.

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

During the time HPCI was inoperable the RCIC system remained available for high pressure injection in the event of an emergency. The Standby Feedwater (SBFW) [SJ] system was also available for high pressure injection. Additionally, the Automatic Depressurization System (ADS) was available to reduce reactor pressure to within the capabilities of the low pressure Emergency Core Cooling Systems (ECCS) [BO][BM]. TS 3.5.1 allows HPCI to be taken out of service for up to 14 days provided that RCIC is available. HPCI is required for design basis accidents such as a loss of coolant accident (LOCA). No accidents or transients requiring HPCI occurred or were in progress during the HPCI inoperability.

Offsite power was operable throughout the event, being supplied to both the 120 kV and 345 kV switchyards from the transmission network by five transmission lines. Division 1 of the onsite power sources (EDGs 11 and 12) remained operable throughout the event. Division 2 EDGs (EDGs 13 and 14) were declared inoperable due to the condition affecting the Division 2 EDG load sequencer. However, further review identified that the condition affecting the load sequencer did not impact operation of the EDGs themselves. In the event of a loss of offsite power, EDGs 13 and 14 would still have started automatically and Division 2 loads would have been shed to prepare to accept the required loads. Due to failure of the load sequencer, loads would not have automatically been added. However, with operator actions to close breakers (two per EDG), many of the ESF loads would then have been manually added as needed. These operator actions are performed from the Main Control Room (MCR) and are identified in existing Fermi 2 procedures which operators are trained to utilize. As a result, EDGs 13 and 14 were considered inoperable but available. In a postulated transient or accident where offsite power and the Division 1 EDGs were both lost, the Division 2 EDGs could still have

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been utilized with operator actions to supply power to required ESF loads.

Based on this discussion, the safety significance of this event is very low. There were no radiological releases associated with this event.

CAUSE OF THE EVENT

The cause of this event was a voltage spike of unknown origin. During initial troubleshooting both the Division 2 EDG Automatic Load Sequencer power supply and HPCI Inverter were found to have a short between the positive and negative input terminals and the inputs remained isolated from the chassis. Initial troubleshooting for the EDG load sequencer found two switching transistors shorted and for the HPCI inverter found five metal -oxide semiconductor field effect transistors (MOSFETs) shorted and a possible capacitor failed.

During the troubleshooting of the charger's cycling voltage it was found that the phase "A" auxiliary transformer was having intermittent problems before it had a hard failure. It was turning on and off causing the silicone-controlled rectifiers (SCRs) to shut off. The voltage output of the transformer should be approximately 45v, but was 0v while the primary side was 120v. The transformer was replaced with a new transformer and the battery charger functioned as expected and was placed back into service. Following further troubleshooting the issue with the battery charger was independent of the Division 2 EDG Automatic Load Sequencer power supply and HPCI inverter issues.

The failure mechanism of the inverter and the sequencer failures were caused by a voltage transient on the DC bus. This transient was created from an unknown source either electrically connected to 2PB2-6 or magnetically coupled to the supply cables of the devices. These findings were independently reviewed by a subject matter expert. A modification will be applied to monitor the input voltage on the circuit that may identify this unknown source of the voltage spike.

CORRECTIVE ACTIONS

A monitoring system is being installed to monitor input voltage on the circuit that will potentially identify a voltage transient which will help identify this unknown source.

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NARRATIVE**PREVIOUS OCCURRENCES**

There have been previous occurrences of unplanned HPCI inoperability in the past several years, for example LERs 2018-005 and 2019-004. However, those occurrences were due to a support system, the Mechanical Draft Cooling Towers (MDCT), not being able to support operability of the Division 2 Emergency Equipment Cooling Water (EECW) system which is required for the operability of the HPCI room cooler. Therefore, the corrective actions taken to resolve those previous occurrences would not have prevented the issue described in this LER.